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Assembly line having carrier for transporting components and method of operation thereof

Abstract

An assembly line having a carrier for transporting components which is structured and dimensioned for receiving components of two different devices, so that the assembly line can be switched from assembling one device to assembling the other device without changing or altering the series of carriers on the assembly line, and a method of operating the assembly line.

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Background/Summary

(1) The present application is a Continuation of U.S. patent application Ser. No. 18/341,981, filed Jun. 27, 2023, entitled “ASSEMBLY LINE HAVING CARRIER FOR TRANSPORTING COMPONENTS AND METHOD OF OPERATION THEREOF,” which issued into U.S. Pat. No. 12,012,294, entitled “ASSEMBLY LINE HAVING CARRIER FOR TRANSPORTING COMPONENTS AND METHOD OF OPERATION THEREOF,” on Jun. 18, 2024, which is a Division of U.S. patent application Ser. No. 17/748,865, filed May 19, 2022, entitled “ASSEMBLY LINE HAVING CARRIER FOR TRANSPORTING COMPONENTS AND METHOD OF OPERATION THEREOF,” which issued into U.S. Pat. No. 11,724,893, entitled “ASSEMBLY LINE HAVING CARRIER FOR TRANSPORTING COMPONENTS AND METHOD OF OPERATION THEREOF,” on Aug. 15, 2023. All of which are incorporated herein by reference.

(1) The present description relates to a carrier for transporting components on an assembly line that is structured and dimensioned for receiving components of two different devices, so that the assembly line can be switched from assembling one device to assembling the other device without changing or altering the series of carriers on the assembly line.

BACKGROUND

(2) In some instances, a single assembly line can be used for assembling different devices whose respective components are similar to each other but have different dimensions and shapes. Switching that assembly line from assembling one device to assembling another can be a time-consuming evolution that idles production and is inefficient. One example is an assembly line that assembles nasal spray devices. Such devices may comprise three components that are assembled together: a filled vial, a vial holder into which the filled vial is inserted, and an actuator grip that is inserted over and surrounds at least part of the vial holder and filled vial. Such devices come in at least two sizes containing different amounts of nasal spray. Switching an assembly line from assembling one size to assembling another is time consuming and inefficient, considering that producing of such devices is measured in hundreds of doses per minute.

SUMMARY

(3) A method of operating an assembly line having assembly stations and a plurality of carriers for transporting components of a vial-based device between the assembly stations is provided. Each carrier comprises first and second device receptacles, the first device receptacle being structured and dimensioned differently than the second device receptacle, the first device receptacle being adapted for receiving components of a first vial-based device, and the second device receptacle being adapted for receiving components of a second vial-based device. The carrier is further adapted for attachment to a car that travels along the assembly line, the plurality of carriers being arranged in a series of cars. The method comprises the steps of assembling a plurality of either one of the first and second vial-based devices; switching the assembly line to assemble the other one of the first and second vial-based devices without changing or altering the series of the carriers on the assembly line; and assembling a plurality of the other one of the first and second vial-based devices.

(4) The first device receptacle may comprise a first vial holder support for receiving a first vial holder of the first vial-based device. The first device receptacle may comprise a first actuator grip support for receiving a first actuator grip of the first vial-based device.

(5) In another example, the first device receptacle comprises a cylindrical tube, having a top that is open wide enough for receipt of a first actuator grip of the first vial-based device, having a diameter and a length structured to support the first actuator grip, and having a bottom with a circular opening that is wide enough for at least part of a first vial holder of the first vial-based device to pass through the opening but narrow enough to prevent the first actuator grip from passing through the opening; and a bottom support located underneath the bottom of the cylindrical tube, and structured to support the at least part of the first vial holder passing through the opening.

(6) In some examples, when the method is in operation, the at least part of the first vial holder passes through the cylindrical tube and through the opening, and is supported by the bottom support and by the bottom of the cylindrical tube; a first filled vial of the first vial-based device is inserted into the first vial holder after the at least part of the first vial holder has passed through the cylindrical tube and through the opening and the at least part of the first vial holder is supported by the bottom support and by the bottom of the cylindrical tube; and the first actuator grip is inserted over and surrounds at least part of the first vial holder and the first filled vial, and is supported by the cylindrical tube, after the first filled vial has been inserted into the first vial holder. The method cylindrical tube may be freely rotatable about its vertical axis to any angular orientation within the first device receptacle.

(7) The second device receptacle may comprise a second vial holder support for receiving a second vial holder of the second vial-based device. The second device receptacle may comprise a second actuator grip support for receiving a second actuator grip of the second vial-based device.

(8) The second device receptacle may comprise: a top support having a length and a width structured to allow a second actuator grip of the second vial-based device to fit over and to be supported by the top support, and having a circular opening that is wide enough for at least part of a second vial holder of the second vial-based device to pass through the opening; and a bottom support located underneath the top support, and structured to support the at least part of the second vial holder passing through the opening.

(9) In some examples of when the method is in operation, the at least part of the second vial holder passes through the opening, and is supported by the bottom support and by the top support; a second filled vial of the second vial-based device is inserted into the second vial holder after the at least part of the second vial holder has passed through the opening and the at least part of the second vial holder is supported by the bottom support and by the top support; and the second actuator grip is inserted over and surrounds the top support and at least part of the second vial holder and the filled second vial, and is supported by the top support, after the second filled vial has been inserted into the second vial holder.

(10) In some embodiments, an assembly line for assembling first vial-based devices and second vial-based devices, the first vial-based devices being dimensioned differently than the second vial-based devices are provided. The assembly line comprises a series of the carriers, a series of the cars, the series of the carriers being attached to the series of the cars, respectively; a series of the assembly stations; and a conveyor system for transporting the cars between the assembly stations.

(11) Each carrier comprises: first and second device receptacles, the first device receptacle being adapted for receiving components of the first vial-based device, and the second device receptacle being adapted for receiving components of the second vial-based device. The carrier is adapted for attachment to a car that travels along the assembly line. When a series of the carriers is attached to a series of the cars, respectively, the assembly line can be switched from assembling either one of the first and second vial-based devices to assembling the other one of the first and second vial-based devices without changing or altering the series of the carriers on the assembly line.

(12) The assembly line may further comprise: a vial holder insertion station for inserting a first or a second vial holder into each of the carriers; a filled vial insertion station for inserting a first or a second filled vial into the vial holder that has been inserted into each of the carriers; and an actuator grip insertion station for inserting a first or a second actuator grip over and surrounding at least part of the vial holder and filled vial that has been inserted into each of the carriers. The components of the first vial-based device comprise the first vial holder, the first filled vial, and the first actuator grip. The components of the second vial-based device comprise the second vial holder, the second filled vial, and the second actuator grip. The series of assembly stations comprise the vial holder insertion station, the filled vial insertion station, and the actuator grip insertion station.

(13) The series of assembly stations may further comprise a packaging directing station for directing the assembled first and the second vial-based devices to a selected station for packaging.

(14) The assembly line may further comprise a pull-test mechanism that has a first part for gripping an actuator grip of at least one of the first and second vial-based devices after assembled, and a second part for gripping a vial holder of the least one of the first and second vial-based devices after assembled, the first and second parts being connected by a spring and being operated by temporarily pulling them apart to a specified tension to test whether the actuator grip is secured to the vial holder.

(15) The assembly line may further comprise: a series of individual mover pucks; each individual mover puck being a holder into which either the assembled first vial-based device or the assembled second vial-based device is placed for transport along a conveyor belt; where each individual mover puck being structured and dimensioned to accommodate either the assembled first vial-based device or the assembled second vial-based device, whichever is required.

(16) The assembly line may further comprise a labeling station, wherein the labeling station comprises at least a first labeling module capable of applying wrap-around labels when assembling at least one of the first and second vial based devices and applying spot labels when assembling the other of the first and second vial based devices. The labeling station may further comprise a second labeling module capable of applying spot labels when assembling the other of the first and second vial-based devices.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The following drawings illustrate the concepts of the present invention. Illustrations of exemplary embodiments are not necessarily drawn to scale.
- (2) FIG. 1 illustrates exemplary components of a Bi-Dose device and of a Unit-Dose device.
- (3) FIG. 2 shows a perspective view of an example of a carrier for transporting components of a vial-based device between assembly stations on an assembly line.

- (4) FIG. 3 shows a front view of the carrier of FIG. 2.
- (5) FIG. 4 shows a rear view of the carrier of FIG. 2.
- (6) FIG. 5 shows a right-side view of the carrier of FIG. 2.
- (7) FIG. 6 shows a left side view of the carrier of FIG. 2.
- (8) FIG. 7 shows a top view of the carrier of FIG. 2.
- (9) FIG. 8 shows a bottom view of the carrier of FIG. 2.
- (10) FIG. 9 illustrates an embodiment of a carrier attached to a car.
- (11) FIG. 10A illustrates an example insertion of a vial holder into the carrier attached to a car.
- (12) FIG. 10B illustrates an example insertion of a filled vial into the carrier attached to a car.
- (13) FIG. 10C illustrates an example insertion of an actuator grip into the carrier attached to a car.
- (14) FIG. 10D illustrates an example of an assembled device in the carrier attached to a car.
- (15) FIG. 11A illustrates an example insertion of a different vial holder into the carrier attached to a car.
- (16) FIG. 11B illustrates an example insertion of a different filled vial into the carrier attached to a car.
- (17) FIG. 11C illustrates an example insertion of a different actuator grip into the carrier attached to a car.
- (18) FIG. 11D illustrates an example of a different assembled device in the carrier attached to a car.
- (19) FIG. 12 is a schematic representation of an exemplary assembly line using the carrier of FIG. 2.
- (20) FIG. 13 illustrates an example of a pull-test mechanism.
- (21) FIG. 14A illustrates an exemplary mover puck.
- (22) FIG. 14B illustrates the mover puck of FIG. 14A occupied by an assembled Bi-Dose device.
- (23) FIG. 14C illustrates the mover puck of FIG. 14A occupied by an assembled Unit-Dose device.

DETAILED DESCRIPTION

- (24) FIG. 1 illustrates exemplary components of a Bi-Dose device: vial holder **101**, filled vial **103**, and actuator grip **105**; and of the Unit-Dose device: vial holder **111**, filled vial **113**, and actuator grip **115**. Actuator grip **105** may be modified to include additional area for engagement with a user's fingers during use. Advantageously, an assembly line using the carriers described and claimed in the present application can assemble **200** units per minute, without lengthy downtimes for switching from assembling a Bi-Dose device to assembling a Unit-Dose device and vice versa.
- (25) FIGS. 2-8 show perspective, front, rear, right side, left side, top and bottom views, respectively, of an example of a carrier **200** for transporting components of a vial-based device between assembly stations on an assembly line. The carrier **200** is adapted for attachment to a car **230** designed to be transported along an assembly line **300** (discussed below). For example, a carrier **200** can be bolted to a car **230** through openings **220** as seen in FIGS. 2-4. An embodiment of a carrier **200** attached to a car **230** is illustrated in FIG. 9.
- (26) The illustrated carrier **200** includes a receptacle seen on the left side of FIG. 3, for example, which is structured and dimensioned for receiving components of the Bi-Dose device illustrated in FIG. 1. Carrier **200** includes a receptacle seen on the right side of FIG. 3, for example, which is structured and dimensioned for receiving components of the Unit-Dose device illustrated in FIG. 1.
- (27) The receptacle seen on the left side of FIG. 3 includes a sleeve **201** surrounding a cylindrical tube having a top **202** that is open wide enough for receipt of an actuator grip **105**, having a diameter and a length structured to support an actuator grip **105**, and having a bottom **203** with a circular opening **204** that is wide enough for at least part of a vial holder **101** to pass through the opening **204** but narrow enough to prevent an actuator grip **105** from passing through the opening **204**. The receptacle seen on the left side of FIG. 3 also includes a bottom support **206**, located underneath the bottom **203** of the sleeve **201**, that is structured to support at least part of a vial holder **101** passing through the opening **204**.
- (28) As a result, at least part of a vial holder **101** can pass through the cylindrical tube and through

the opening **204** and can be supported by the bottom support **206** and by the bottom **203** of the cylindrical tube. Then, a filled vial **103** can be inserted into the vial holder **101**. Then, an actuator grip **105** can be inserted over and can surround at least part of the vial holder **101** and the filled vial **103** and can be supported by the cylindrical tube.

(29) FIGS. **10A-10D** illustrate an example of the sequential insertion of a vial holder **101** (FIG. **10A**), insertion of a filled vial **103** (FIG. **10B**), insertion of an actuator grip **105** (FIG. **10C**), and an assembled Bi-Dose device in the receptacle seen on the left side of FIG. **10D**.

(30) In the illustrated embodiment, the combination of the bottom support **206** and the bottom **203** of the cylindrical tube constitute a vial holder support, and the cylindrical tube constitutes an actuator grip support. Many embodiments can be devised, as long as they are structured and dimensioned for receiving and supporting components of a device being assembled along an assembly line.

(31) In some embodiments, the cylindrical tube is freely rotatable about its vertical axis to any angular orientation within the receptacle seen on the left side of FIG. **3**. For example, it can rotate freely within sleeve **201**. For example, this ability to rotate facilitates the application of a wrap-around label to an assembled device. The label can be applied at one point, and then the device can be rotated 360° while applying pressure via a belt to ensure that the label adheres to the device.

(32) The receptacle seen on the right side of FIG. **3** includes a top support **211** having a length and a width structured to allow an actuator grip **115** to fit over and to be supported by the top support **211**, and having a circular opening **212** that is wide enough for at least part of a vial holder **111** to pass through the opening **212**. The receptacle seen on the right side of FIG. **3** also includes a bottom support **214**, located underneath the top support **211**, and structured to support at least part of the vial holder **111** passing through the opening **212**.

(33) As a result, at least part of a vial holder **111** can pass through the opening **212** and can be supported by the bottom support **214** and by the top support **211**. Then, a filled vial **113** can be inserted into the vial holder **111**. Then, an actuator grip **115** can be inserted over and can surround the top support **211** and at least part of the vial holder **111** and the filled vial **113** and can be supported by the top support **211**.

(34) FIGS. **11A-11D** illustrate an example of the sequential insertion of a vial holder **111** (FIG. **11A**), insertion of a filled vial **113** (FIG. **11B**), insertion of an actuator grip **115** (FIG. **11C**), and an assembled Unit-Dose device in the receptacle seen on the right side of FIG. **11D**.

(35) In the illustrated embodiment, the combination of the bottom support **214** and the top support **211** constitute a vial holder support, and the top support **211** constitutes an actuator grip support. Many embodiments can be devised, as long as they are structured and dimensioned for receiving and supporting components of a device being assembled along an assembly line.

(36) FIG. **12** is a schematic representation of an exemplary assembly line **300** for assembling both the Bi-Dose and the Unit-Dose devices whose components are illustrated in FIG. **1**. A series of carriers **200** is attached to a series of cars **230**, respectively, forming a series of car/carriers such as those illustrated in FIG. **9**. The car/carriers are transported along a conveyor system **302** between a series of assembly stations including, for example, a vial holder insertion station **304**, a filled vial insertion station **306**, an actuator grip insertion station **308**, and a packaging directing station **310**.

(37) In the illustrated example, vial holder feeder **312** is loaded with vial holders selected for the device being assembled and sends the vial holders to vial holder insertion station **304** where a vial holder is inserted into the applicable receptacle of each carrier **200**. Filled vial feeder **314** is loaded with filled vials selected for the device being assembled and sends the filled vials to filled vial insertion station **306** where a filled vial is inserted into the vial holder that has been inserted into each carrier **200**. Actuator grip feeder **316** is loaded with actuator grips selected for the device being assembled and sends the actuator grips to actuator grip insertion station **308** where an actuated grip is inserted over and surrounds at least part of each vial holder/filled vial that has been inserted into each carrier **200**. Label feeder **318** is loaded with labels selected for the device being

assembled and sends the labels to packaging directing station **310** where a label is applied to each assembled device and the device is directed to a selected packaging station **320A**, **320B**, or **320C**, for example.

(38) In some embodiments, the packaging directing station **310** comprises a labeling station having a plurality of labeling modules. In one example, the labeling station comprises a labeling module that is capable of applying wrap-around labels and spot labels. In the case of the Bi-Dose device, as set forth above, the receptacle allows for rotation of an assembled device to facilitate application of a wrap-around label. A first labeling module includes, for example, two servo-driven conveyor belts to rotate the Bi-Dose device at a speed synchronized with the carrier travel to apply a wrap-around label. For devices that are not amenable to rotation, front and back labels may be applied. In one example, a second labeling module is added. For example, at the first labeling module, a first side label (e.g., back of the device) may be applied to a Unit Dose device, and at the second labeling module, a second side label (e.g., front) may be applied to a Unit Dose device. The labeling modules may include a laser to write data onto labels and a vision system to optically inspect labels before application to a device. In this way, different labeling requirements can be accomplished without change-over of any equipment on the assembly line.

(39) Because each carrier **200** can be used in assembling either a Bi-Dose device or a Unit-Dose device in the described example, carriers **200** attached to cars **230** can be transported along conveyor system **302** without change or alteration when the assembly line is switched from assembling the Bi-Dose device to assembling the Unit-Dose device, and vice versa.

(40) When switching from assembling one device to assembling the other, the applicable components and labels are fed in bulk to the vial holder feeder **312**, the filled vial feeder **314**, the actuator grip feeder **316** and the label feeder **318**, respectively. However, same carriers **200** continue to be transported along the conveyor system **302**.

(41) In some embodiments, multiple carriers **200** move together from one assembly station **304**, **306**, **308** and **310** to the next. For example, each assembly station **304**, **306**, **308** and **310** can be designed to accommodate eight carriers **200** simultaneously. When the function of each of the assembly stations **304**, **306**, **308** and **310** is completed, all eight carriers **200** at each of the assembly stations **304**, **306** and **308** are transported to the next assembly station **306**, **308** and **310**, the empty carriers **200** at packaging directing station **310** move on, and a new set of eight empty carriers **200** arrive at vial holder insertion station **304**. In some embodiments, the conveyor system **302** uses electro-magnetic pulses to transport the cars **230** between the assembly stations **304**, **306**, **308** and **310**.

(42) Various features specific to the devices being assembled are built into the assembly line **300**. For example, there can be quality control tests to confirm the integrity and operability of a component. In some embodiments, the assembly line **300** can include a pull-test mechanism **330** such as illustrated in FIG. 13. The test mechanism **330** has a first part **331** for gripping an actuator grip **115** of at least the Unit-Dose devices after assembled, and a second part **332** for gripping a vial holder **111** of at least the Unit-Dose devices after assembled. The first and second parts **331** and **332** are connected by a spring and are operated by temporarily pulling them apart to a specified tension to test whether the actuator grip **115** is secured to the vial holder **111**. Typically, vial holder **111** has a snap ring **112** for retaining vial holder **111** within actuator grip **115**. The first part **331** has a cup for holding a top of actuator grip **115**, the second part **332** has two legs for holding vial holder **111** and pull-test mechanism **330** detects whether the snap ring **112** has been damaged and is not performing its function.

(43) The following are examples of some other features included in the assembly line **300** in some embodiments. For example, the assembly line **300** can include reference mechanisms to ensure the correct spacing of carriers **200** at the assembly stations **304**, **306**, **308** and **310**. There can be sensors to ensure that the carriers **200** are empty when they arrive at vial holder insertion station **304**. There can be pick-and-place A robots at each of the assembly stations **304**, **306**, **308** and **310** for accurate

insertion of components into the applicable receptacle for the device being assembled, or for transfer of assembled devices from the receptacles to mover pucks **340** (discussed below).

(44) The alternative packaging stations **320A**, **320B**, or **320C**, for example, accommodate not only the two different devices, but also allow for different types of packaging such as for individual devices or for different numbers of multiple devices, and allow for the use of different types of packaging materials. The packaging directing station **310** can be controlled to direct certain numbers of devices to the selected packaging stations **320A**, **320B**, and **320C**, for example.

(45) FIG. **14A** is an illustration of an exemplary mover puck **340** structured and dimensioned so that any individual mover puck **340** can accommodate either an assembled Bi-Dose device or an assembled Unit-Dose device, whichever is required. FIG. **14B** illustrates an assembled Bi-Dose device placed in a mover puck **340**. FIG. **14C** illustrates an assembled Unit-Dose device placed in a mover puck **340**. The mover pucks **340** are holders into which an assembled device can be placed for transport along a conveyor belt. For example, individual assembled devices that are removed from carriers **200** at packaging directing station **310** can be placed in respective mover pucks **340** for transport to a selected packaging station **320A**, **320B**, or **320C**, for example.

(46) It will be understood that the disclosed carrier and assembly line device can be modified without departing from the teachings of the invention. Accordingly, the scope of the invention is to be limited only as necessitated by the accompanying claims.

Claims

1. A method of operating an assembly line having assembly stations and a plurality of carriers for transporting components of a vial-based device between the assembly stations, each carrier comprising first and second device receptacles, the first device receptacle being adapted for receiving components of a unit dose vial-based device, and the second device receptacle being adapted for receiving components of a bi-dose vial-based device, and the carrier being adapted for attachment to a car that travels along the assembly line, the plurality of carriers being arranged in a series of cars, the method comprising: assembling a plurality of either one of the unit dose and bi-dose vial-based devices; switching the assembly line to assemble the other one of the unit dose and bi-dose vial-based devices without changing or altering the series of the carriers on the assembly line; and assembling a plurality of the other one of the unit dose and bi-dose vial-based devices.
2. The method according to claim 1, wherein the first device receptacle comprises: a cylindrical tube, having a top that is open wide enough for receipt of a first actuator grip of the unit dose vial-based device, having a diameter and a length structured to support the first actuator grip, and having a bottom with a circular opening that is wide enough for at least part of a first vial holder of the unit dose vial-based device to pass through the opening but narrow enough to prevent the first actuator grip from passing through the opening; and a bottom support located underneath the bottom of the cylindrical tube and structured to support the at least part of the first vial holder passing through the opening.
3. The method according to claim 2, wherein, in operation, the at least part of the first vial holder passes through the cylindrical tube and through the opening, and is supported by the bottom support and by the bottom of the cylindrical tube; a first filled vial of the unit dose vial-based device is inserted into the first vial holder after the at least part of the first vial holder has passed through the cylindrical tube and through the opening and the at least part of the first vial holder is supported by the bottom support and by the bottom of the cylindrical tube; and the first actuator grip is inserted over and surrounds at least part of the first vial holder and the first filled vial, and is supported by the cylindrical tube, after the first filled vial has been inserted into the first vial holder.
4. The method according to claim 2, wherein the cylindrical tube is freely rotatable about its vertical axis to any angular orientation within the first device receptacle.

5. The method according to claim 1, wherein the second device receptacle comprises: a top support having a length and a width structured to allow a second actuator grip of the bi-dose vial-based device to fit over and to be supported by the top support, and having a circular opening that is wide enough for at least part of a second vial holder of the bi-dose vial-based device to pass through the opening; and a bottom support located underneath the top support, and structured to support the at least part of the second vial holder passing through the opening.
 6. The method according to claim 5, wherein, in operation, the at least part of the second vial holder passes through the opening, and is supported by the bottom support and by the top support; a second filled vial of the bi-dose vial-based device is inserted into the second vial holder after the at least part of the second vial holder has passed through the opening and the at least part of the second vial holder is supported by the bottom support and by the top support; and the second actuator grip is inserted over and surrounds the top support and at least part of the second vial holder and the filled second vial, and is supported by the top support, after the second filled vial has been inserted into the second vial holder.
 7. The method according to claim 1, wherein the first device receptacle comprises a first vial holder support for receiving a first vial holder of the unit dose vial-based device.
 8. The method according to claim 1, wherein the first device receptacle comprises a first actuator grip support for receiving a unit dose actuator grip of the first vial-based device.
 9. The method according to claim 1, wherein the second device receptacle comprises a bi-dose vial holder support for receiving a second vial holder of the second vial-based device.
 10. The method according to claim 1, wherein the second device receptacle comprises a second actuator grip support for receiving a second actuator grip of the bi-dose vial-based device.
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